Ch 6.3: PCR

Lecture 20 - CMSE 381

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Fri, Oct 18, 2024

Announcements

Last time:

- PCA

This lecture:

- PCR

Announcements:

- Exam #2 on Friday!
  - Bring 8.5x11 sheet of paper
  - Handwritten both sides
  - Anything you want on it, but must be your work
  - You will turn it in
  - Non-internet calculator if you want it

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Subset selection	6.1	HW #4 Due Weds 10/9
17	Fri	10/11	Shrinkage: Ridge	6.2.1	
18	Mon	10/14	Shrinkage: Lasso	6.2.2	
19	Wed	10/16	Dimension Reduction	6.3	
20	Fri	10/18	Overflow, Possibly more dimension reduction?		HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		

Section 1

Last time

Shrinkage

Diff formulation!

Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to:

Least Squares:

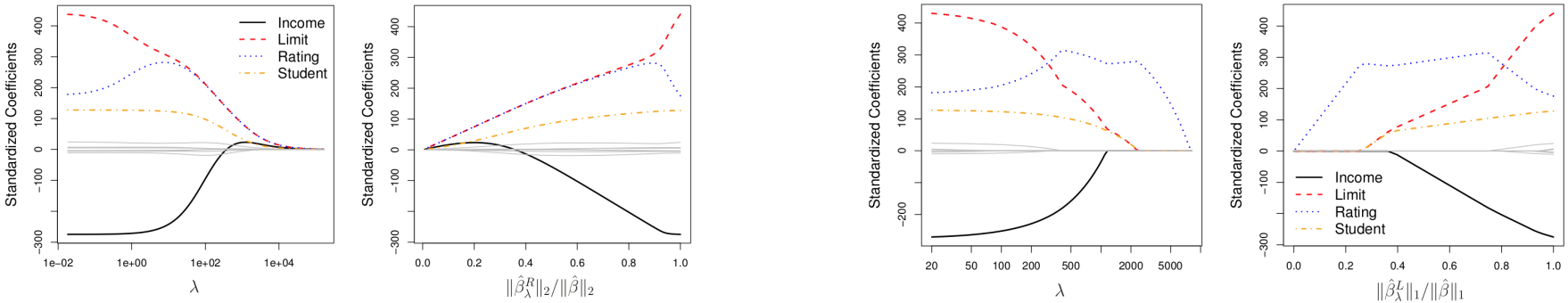
No constraints

Ridge:

$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

The Lasso:

$$\sum_{j=1}^{p} |\beta_j| \leq s$$



Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

$M \ll P$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

$$Z_m = \varphi_{1m} X_1 + \varphi_{2m} X_2 + \cdots + \varphi_{pm} X_p$$

The goal:

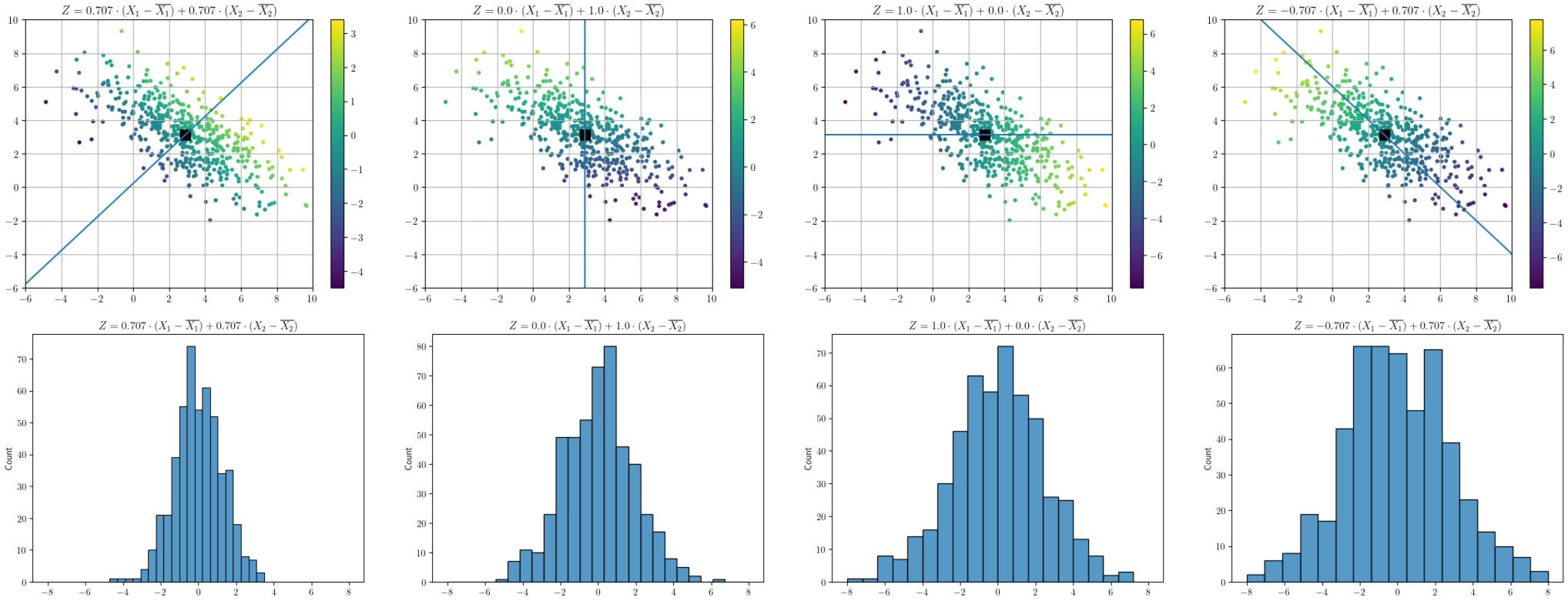
- Find good φ's (PCA)
- Fit regression model on Z_i's using least squares (PLS)

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

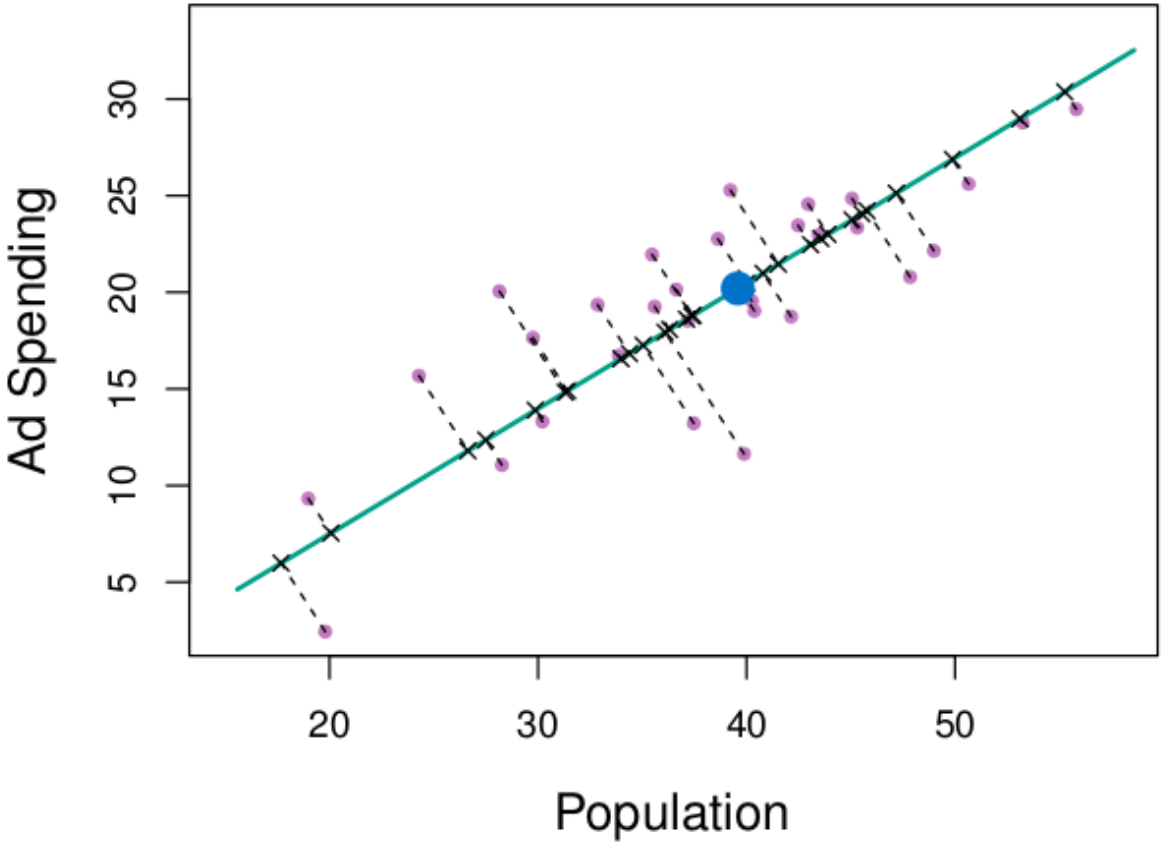
previously β_i

- Hope that lower dimensions means less overfitting

PCA - First PC

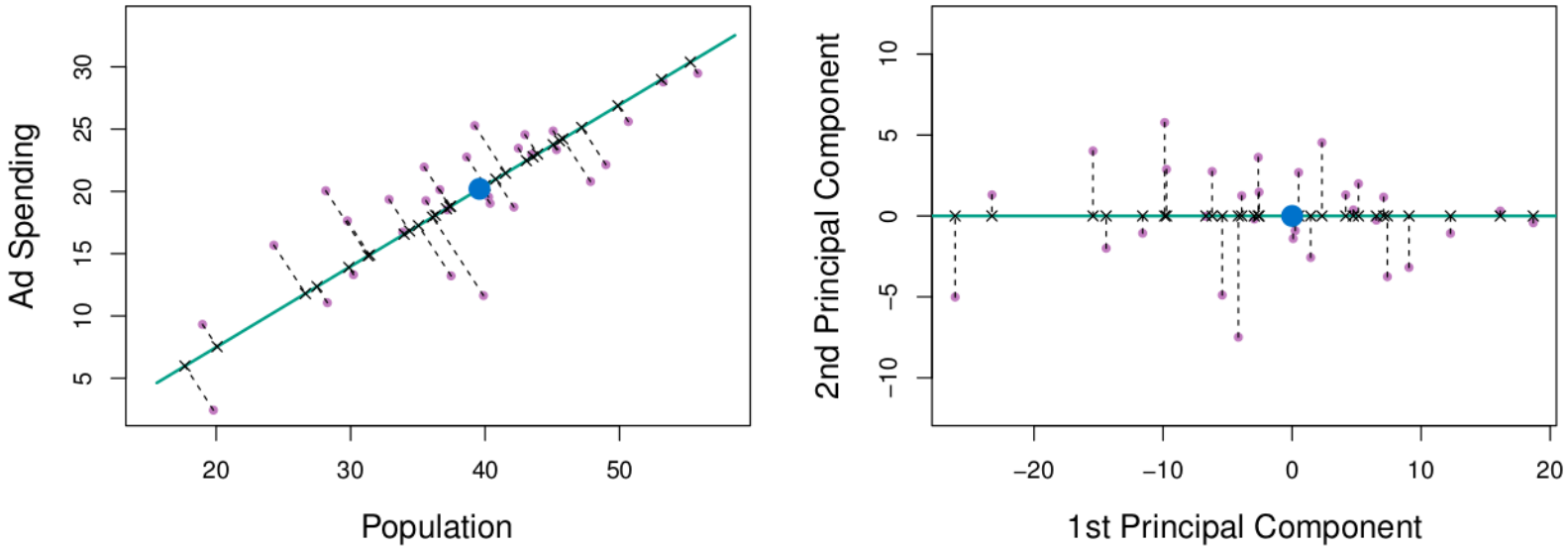

$Z = 0.707 \cdot (X_1 - \overline{X_1}) + 0.707 \cdot (X_2 - \overline{X_2})$
$Z = 0.0 \cdot (X_1 - \overline{X_1}) + 1.0 \cdot (X_2 - \overline{X_2})$
$Z = 1.0 \cdot (X_1 - \overline{X_1}) + 0.0 \cdot (X_2 - \overline{X_2})$
$Z = -0.707 \cdot (X_1 - \overline{X_1}) + 0.707 \cdot (X_2 - \overline{X_2})$
Count

Projection onto first PC



$$Z_1 = 0.839 \cdot (\text{pop} - \overline{\text{pop}}) + 0.544 \cdot (\text{ad} - \overline{\text{ad}})$$

Drawing points in PC space



Section 2

Principal Components Regression

So you've found your PCA coefficients

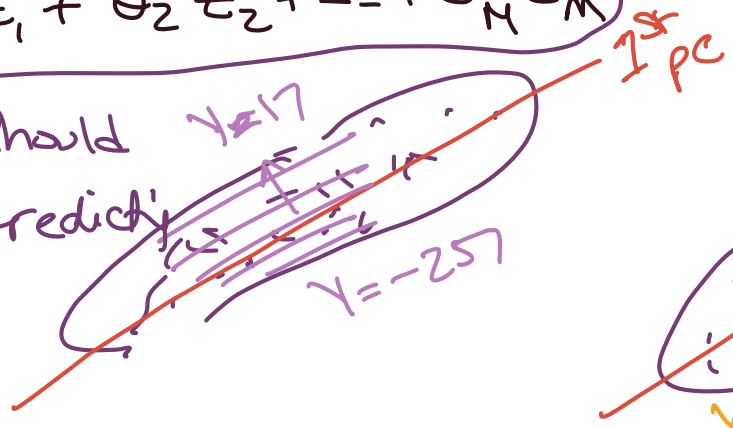
Now what?

Do linear regression on $Z_1, \ldots, Z_M$

Linear model

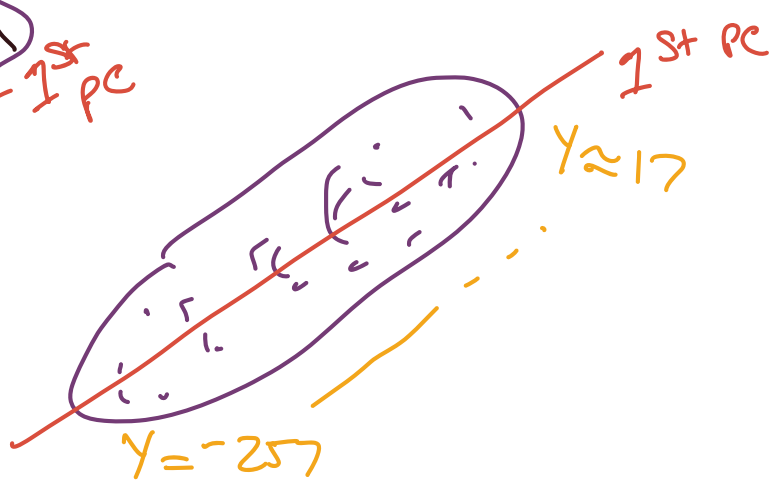
$$Y = \theta_0 + \theta_1 Z_1 + \theta_2 Z_2 + \cdots + \theta_M Z_M$$

If this is true, should do a better job predicting on Z_i's instead of the X_i's.



What are we assuming?

The directions in which $X_1, \ldots, X_p$ have the most variation are also the ones associated to Y.



Interpretation of PCR coefficients

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

Learned model:

$$y = \theta_0 + \theta_1 Z_1 + \cdots + \theta_M Z_M$$

ex) $p = 3$, $M = 2$ $\quad \varphi = \begin{pmatrix} 1 & -1 \\ 3 & 1/2 \\ 0 & 1/2 \end{pmatrix}$

Learned model $Y = 2Z_1 - Z_2$

Go back to X_i's

$$y = 2(X_1 + 3X_2 + 0X_3) - (-X_1 + \tfrac{1}{2}X_2 + \tfrac{1}{2}X_3)$$

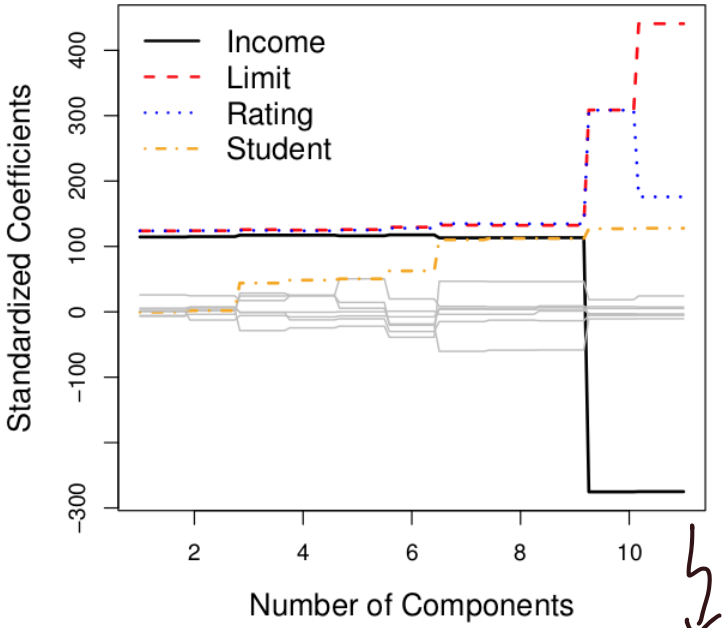
$$= 3X_1 + 2.5X_2 - 0.5X_3$$

Conclusion: Cannot make interpretations of X_i variables using the coef of the Z_i's

Picking M

coef of X_i after converting back to original inputs

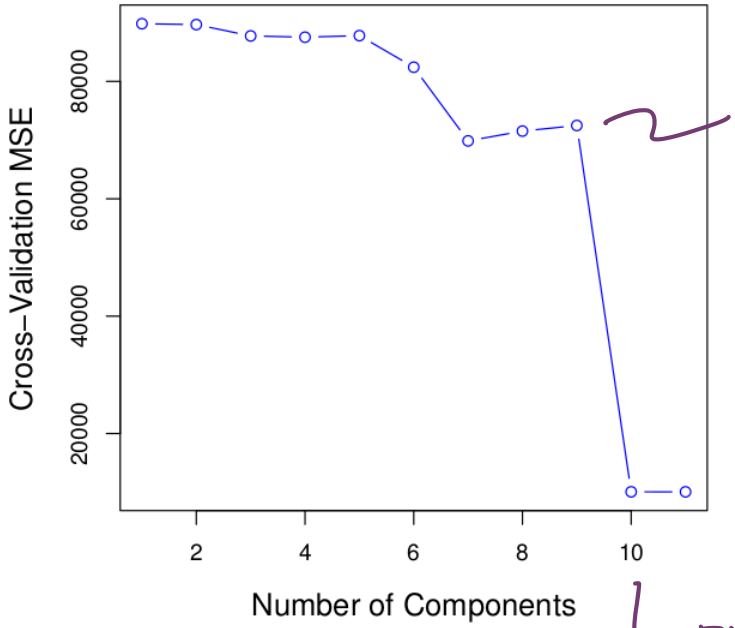
$p = 11$



M

all 11 variables

this is linear regression



way worse @ $M = 9$ components

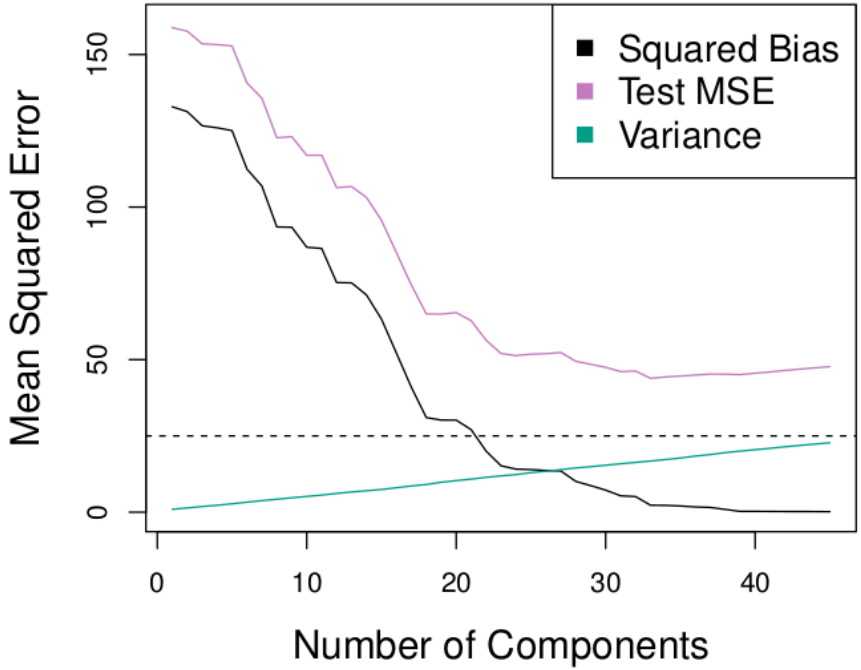
pick $M = 10$

Do PCR with hitters data

Doing better

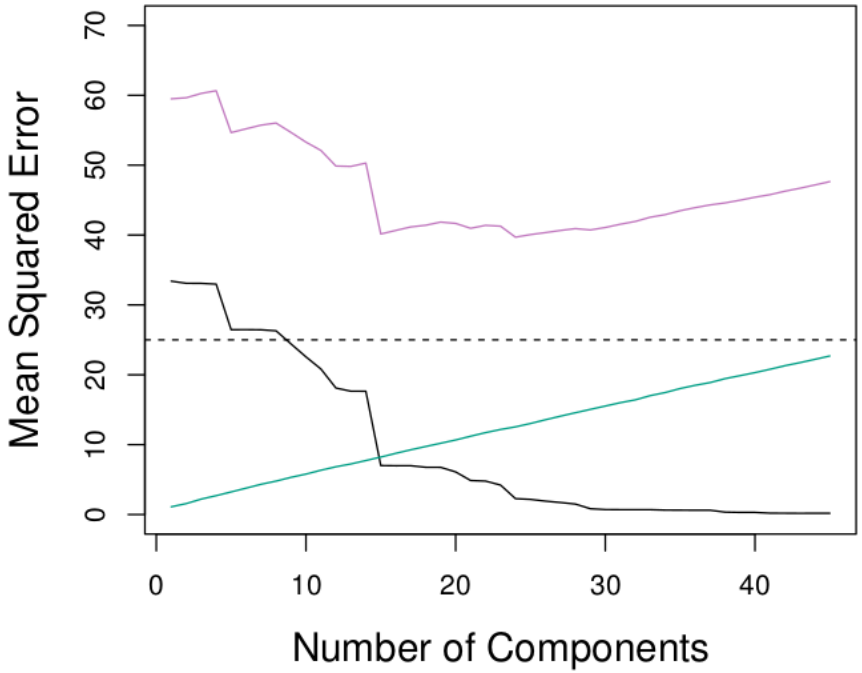
Example with simulated data: $n = 50$ observations of $p = 45$ predictors

Y is a function of **2 predictors**



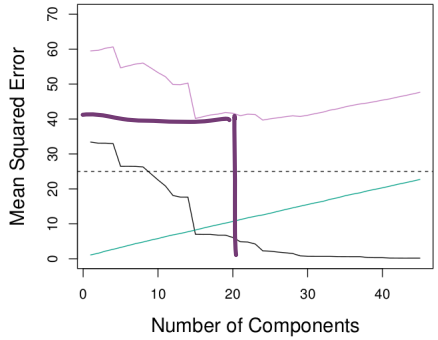
Doing better

Example with simulated data: $n = 50$ observations of $p = 45$ predictors
Y is a function of **all predictors**

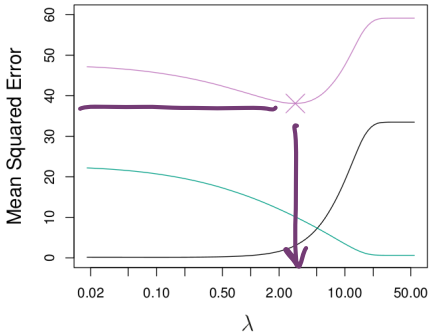


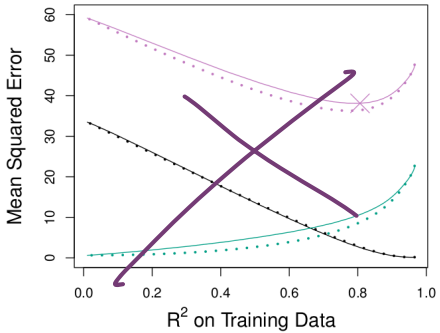
Comparison to results on shrinkage

Y is a function of all predictors



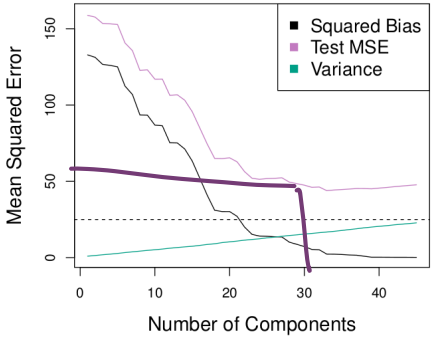
Similar MSE for Ridge / PCR

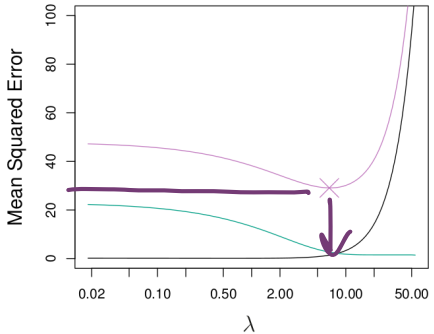


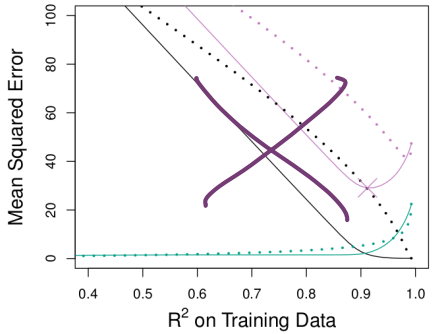


Ridge

Y is a function of 2 predictors







Ridge lower MSE

Properties of PCR

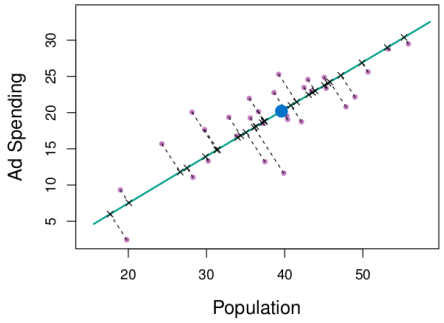
- Standardization is important
  - PCA goes w/ direction of highest variance, so large values skew the results.

- Not a feature selection method
  - $\Rightarrow$ never only using a subset of the inputs

TL;DR

PCR

- Unsupervised dimensionality reduction + linear regression
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



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